

# No Pain, Same Gain: Interference-Gated Aggregation for Constant-Memory Recurrence

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## Abstract

The Transformer established that fixed-state recurrence is dispensable for sequence modeling—*attention is all you need*—but attention pays for that generality with a key-value cache that grows with context (memory  $\propto T$ ,  $O(T^2)$  per-step cost). Growing-memory RNNs [Behrouz et al., 2026] reclaim recurrence by caching one associative memory per segment so capacity grows with length, yet a faithful implementation collapses on multi-query associative recall (MQAR): 25.2% at 16 key-value pairs versus 97.5% for a *fixed* linear-attention memory of the same width. We trace the collapse to *aggregation*, not routing: normalizing each cached segment’s readout before mixing destroys the global-denominator suppression that makes linear attention work, so any non-saturated gate mixes full-scale garbage at signal magnitude. We propose **Interference-Gated Aggregation (IGA)**, which gates *unnormalized* per-segment numerators and denominators inside a single global ratio; with gates initialized to one, IGA is numerically identical to the fixed memory (verified to  $5 \times 10^{-7}$ ), so the fixed memory is a *floor* and collapse is impossible by construction. We pair it with a **tensor-sketch polynomial router** that routes by *learned identity* without reading token IDs—a differentiable relaxation of a count-min sketch—giving recall that is flat in evaluation length even when queries never repeat keys. We then make the recurrent state **constant in sequence length**: a rank- $\leq S$  factorization removes the  $d^2$  term exactly, and capping live segments with a low-rank compressed background bounds growth—0.39 MB at  $d = 64$  versus a KV cache that grows  $\propto T$ , below it from  $T \approx 1\text{k}$  and  $15\times$  smaller projected to a 1T-parameter model, at no recall loss on the headline task. On the literature’s own benchmark—the official Zoology MQAR harness, hardest standard setting (512, 64)—IGA reaches  $99.2 \pm 0.1$  versus Mamba  $23.2 \pm 14.0$  and Based  $79.5 \pm 1.4$ , matching full softmax attention (100.0). On a real character-level language model (TinyStories, 2 layers,  $d = 128$ , 3 seeds, two contexts) IGA matches a fixed-memory baseline of the same width in bits-per-character within seed noise—*constant memory at no perplexity cost*—while a cost analysis shows the one-time training premium is amortized by per-query inference savings that grow with context, since IGA is  $O(T)$  per step with a constant state where softmax attention is  $O(T^2)$  per step with a  $\propto T$  cache. **No pain, same gain**: the cost of growing memory is removed, the quality is retained.

## 1 Introduction

“Attention is all you need” [Vaswani et al., 2017] retired fixed-state recurrence for sequence modeling, but the price is written into the mechanism: a key-value cache that holds every past token. The cache grows linearly with context and the per-step attention product is quadratic, so memory and compute both scale with the sequence. This is the *pain* of the attention paradigm—bearable at short context, dominant at long context, and eventually the binding constraint.

Growing-memory RNNs [Behrouz et al., 2026] offer the opposite bet: keep a recurrent state, but let it *grow* by caching one associative memory per segment of  $S$  tokens and aggregating the cached readouts with a learned gate. Capacity then scales with length while per-token cost stays sub-quadratic. The promise is attention’s recall with recurrence’s cost structure. We find that a faithful single-layer implementation of this promise does not merely fail to beat the fixed memory it generalizes—it collapses far below it (Section 4, Table 1).

This paper makes a single, sharp proposal: a growing-memory model that carries **no pain, same gain**—no growing-memory cost, and the same quality as the strong baselines. We deliver it in three steps. First, we fix the aggregation: the collapse is caused by per-segment readout normalization, and gating the *unnormalized* per-segment state inside one global ratio (IGA) removes it, with the fixed memory as a provable initialization floor. Second, we make the remaining bottleneck—segment *routing*—solvable without token IDs, via a tensor-sketch polynomial router that routes by learned identity. Third, we make the state *constant in sequence length*: a rank- $\leq S$  factorization removes the quadratic-in-width term exactly, and capping live segments with a low-rank compressed background caps growth, giving a constant sub-KV state at no recall loss on the headline task. The tagline names exactly what the measurements support: *no pain*—the growing memory and the  $O(T^2)$  step are gone, leaving a constant state and  $O(T)$  per-step compute; *same gain*—recall matches full attention on the literature’s hardest standard task (99.2 vs 100.0) and bits-per-character matches the fixed-memory baseline on real text (within seed noise). The title is a compression of the result, not a slogan.

## Contributions.

1. **Diagnosis and aggregation fix (§3.1).** The growing-memory collapse is an *aggregation* failure: per-segment normalization erases the match-mass difference between segments. IGA gates unnormalized numerator/denominator pairs inside one global ratio; at gate  $\approx 1$  (init) the model is the fixed memory to  $5 \times 10^{-7}$ , so learning can only *remove* cross-segment interference—the source of the capacity that makes  $K$  small memories beat one large memory.
2. **Learned identity routing (§3.2).** Summed-similarity routing is kernel-family limited: it tracks the fixed memory’s saturation across score families and even under auxiliary supervision, whereas identity-preserving summaries route exactly but exploit verbatim key repetition. Our tensor-sketch polynomial router [Pham and Pagh, 2013] is a differentiable, ID-free relaxation of a count-min sketch that preserves the flat-recall property, and remains flat on **MQAR-indirect**, a benchmark where queries never repeat keys—recovered label-free by distilling a quadratic teacher trained only on the recall loss.
3. **Constant-memory state (§3.3).** A rank- $\leq S$  factorization of the per-segment cache removes the  $d^2$  term exactly ( $< 5 \times 10^{-7}$ ); capping live segments at  $K_{\max}$  with a low-rank compressed background makes the recurrent state constant in  $T$ —0.39 MB at  $d = 64$ , below the KV cache from  $T \approx 1\text{k}$  and  $15\times$  smaller projected to 1T parameters, with no recall loss on the (512, 64) task.
4. **Real-text validation and deployment cost (§5–§6).** On a real character-level language model (3 seeds, two contexts) IGA matches a fixed-memory baseline of the same width in bits-per-character within seed noise—the constant-memory state carries no perplexity penalty—while the associative-recall advantage is shown to be task-specific (it does not transfer to a planted-fact probe on natural text at this scale). A cost analysis shows the one-time training premium (the gate) is amortized by per-query inference savings that grow with context, since IGA is  $O(T)$  per step with a constant state against softmax attention’s  $O(T^2)$  per step with a  $\propto T$  cache.

We position the recall claims at small scale (single layer,  $d = 64$ , synthetic recall) and compare against exact attention, Based [Arora et al., 2024] and Mamba [Gu and Dao, 2023] under the official Zoology protocol [Arora et al., 2023] with 3 seeds throughout; the language-model experiment (§5) is the one real-text setting.

## 2 Background and setup

**MQAR.** Input is  $k_1 \ v_1 \ \dots \ k_N \ v_N \ \text{SEP} \ k_{q1} \ v_{q1} \ \dots \ k_{qQ} \ v_{qQ}$ ; the model must predict each queried value as the next token after its query key. We use the Zoology-style formulation with  $N$  up to 128 pairs,  $Q = 4$  queries, key vocabulary 192, value vocabulary 16 (chance 6.25%); training always at  $N = 48$ , 4000 steps, batch 256, AdamW  $3 \times 10^{-4}$ , identical seed before each model; evaluation at  $N \in \{48, 64, 96, 128\}$  with no retraining. In §4 we additionally use the *official* Zoology generator and training protocol (vocabulary 8192, power-law query gaps, learning-rate sweep over  $\text{logspace}(-4, -2, 4)$ , weight decay 0.1, cosine schedule, early stop at 99%).

**Fixed memory (FIXA).** Causal linear attention [Katharopoulos et al., 2020] with feature map  $\varphi(x) = \text{elu}(x) + 1$ :

$$y_t = \frac{\sum_{j \leq t} \varphi(q_t) \cdot \varphi(k_j) v_j}{\sum_{j \leq t} \varphi(q_t) \cdot \varphi(k_j)}.$$

A short depthwise causal convolution ( $k = 3$ ) before the projections lets one layer bind key–value pairs, as in Based/Mamba. Capacity is bounded by  $d^2$ .

**Growing memory (GRM).** Per segment  $i$ :  $M^{(i)} = \sum_j v_j \varphi(k_j)^\top$ ,  $z^{(i)} = \sum_j \varphi(k_j)$ . At query  $t$ , each cached segment contributes a *normalized* readout  $r_i = M^{(i)} q_t / (z^{(i)} \cdot q_t)$ , mixed by a softmax over match masses. This is the mechanism that collapses.

## 3 Method: Interference-Gated Aggregation

### 3.1 Aggregation that cannot collapse

The fixed memory is *exactly* the gate- $\equiv 1$  aggregation of unnormalized per-segment caches:

$$y_{\text{fix}} = \frac{\sum_i M^{(i)} q}{\sum_i z^{(i)} \cdot q}.$$

IGA inserts the gate inside this ratio:

$$y_t = \frac{N_{\text{cur}} + \sum_{i < s} g_i M^{(i)} q_t}{m_{\text{cur}} + \sum_{i < s} g_i z^{(i)} \cdot q_t}, \quad g_i = \sigma(a(\log s_i - \text{ref}) + b),$$

where  $s_i$  is a routing score and  $\text{ref}$  is the mean (similarity scores) or max (count scores) of  $\log s_i$ . With  $b = 3$ ,  $g \approx 0.95$  everywhere at init  $\Rightarrow$  the model *starts* as the fixed memory; the unit test `test_ungated_equals_fixed` verifies the identity to  $5 \times 10^{-7}$ . The gate can only *remove* interference—which is precisely where the capacity of  $K$  small memories over one large memory comes from.

**Why GRM collapses.** Instrumenting the trained GRM at  $n = 16$ : recall 25.2%, routing accuracy of its softmax gate 90.7%, gate mass on the current segment  $\approx 0$ . The killer is that normalized readouts from wrong segments are garbage *at the same magnitude as the signal*; a 0.7/0.3 softmax mixes 30% full-scale garbage. No fix to the softmax can help; the normalization must move *outside* the mixture—which is what IGA does.

### 3.2 Learned identity routing

With aggregation fixed, IGA’s residual gap to oracle is pure routing. The routing problem is a *kernel-family* problem: every summed-similarity score (first/second moment, elementwise-exponential features, learned embeddings with auxiliary supervision) saturates exactly like the fixed memory, whereas *identity-preserving* summaries—the token histogram or its count-min sketch [Cormode and Muthukrishnan, 2005]—route exactly (100%/93%+) at zero parameters and give recall flat in evaluation length, but they read token IDs and MQAR queries repeat keys verbatim. Real language does not.

We need a score  $s_i(q) \approx$  (occurrences of  $q$ ’s identity in segment  $i$ ), cacheable as a per-segment sum, differentiable, and ID-free. For near-orthonormal embeddings  $\hat{e}$ ,  $(\hat{e}_q \cdot \hat{e}_k)^p \rightarrow \delta(q = k)$  as  $p$  grows, and the degree- $p$  polynomial kernel is exactly linearizable:  $(x \cdot y)^p = \langle x^{\otimes p}, y^{\otimes p} \rangle$ , compressible by tensor sketch [Pham and Pagh, 2013] with  $\mathbb{E}[\text{ts}(x) \cdot \text{ts}(y)] = (x \cdot y)^p$  and  $O(1/m)$  variance for unit vectors—*polynomial* cost in sharpness, where Performer-style estimates of a comparably sharp  $\exp(\tau q \cdot k)$  require variance exponential in  $\tau$  [Choromanski et al., 2021]. The router score is

$$s_i = \text{softplus}(w_1)(\hat{e}_q \cdot \sum_j \hat{e}_{k_j}) + \text{softplus}(w_p) \text{med}_J(\text{ts}(\hat{e}_q) \cdot \sum_j \text{ts}(\hat{e}_{k_j})),$$

with three design elements: (i) the **median of  $J = 3$  independent sketches**—the continuous analogue of count-min’s min—lifts routing from 96.6% to 99.8% in a no-training microbenchmark on the exact MQAR segment statistics ( $p = 4$ ,  $m = 256$ ,  $r = 64$ ), where degree-1 alone yields only 39.9–60.8% (the same saturation that limits the fixed memory); (ii) a **degree-1 term for cold start**, since  $\partial(q \cdot k)^p / \partial \text{emb}$  vanishes at orthogonality for  $p \geq 2$ , giving word2vec-style alignment gradients when embeddings must be *learned*, with learnable mixture weights that let training anneal it away; (iii) **count-vs-similarity gate links**—count scores gate on  $\log(\text{clamp}(s))$  with a max reference (hard suppression from init), while learned-poly scores gate on  $\log(\text{softplus}(s))$  and supervise raw scores as logits, because learned scores can be negative at cold start and  $\log(\text{clamp})$  creates a dead gradient zone that destroys readout training (this distinction cost one full failed run, documented as a negative result). Summary cost:  $r + J \cdot m = 832$  floats/segment (count-min: 256; histogram:  $V$ ).

**MQAR-indirect: identity must be learned.** We introduce a benchmark in which the query block uses a *disjoint* query vocabulary, paired to keys by a bijection fixed across the dataset—sequence structure, positions and protocol unchanged. By construction, ID-counting routers receive zero signal. Two training regimes: (a) auxiliary routing cross-entropy with bijection labels—*privileged* supervision, an upper bound for the kernel family; (b) *distillation*—a minimal quadratic softmax-attention teacher ( $d = 64$ , 1 layer, own embeddings) trained only on the recall loss, whose per-segment attention mass is distilled into the router by KL; the teacher is discarded at inference (quadratic cost only during training, mirroring SMT [Kumar and Isola, 2026]). The teacher solves the task at 100%, confirming softmax attention does not suffer the perverse equilibrium.

### 3.3 Constant-memory state

A growing memory that beats the fixed memory on recall but grows without bound does not yet answer the paradigm question—its state would eventually exceed the KV cache it is meant to replace.

We make the recurrent state **constant in  $T$**  in two steps, both presented as design rather than patch.

**Factored state.** The per-segment cache  $M^{(i)} = \sum_{j \in \text{seg}_i} v_j \varphi(k_j)^\top$  is a sum of  $S$  rank-1 updates, so  $\text{rank}(M^{(i)}) \leq S = 16$ . We factor it exactly as  $M^{(i)} = U^{(i)\top} W^{(i)}$  with  $U^{(i)}, W^{(i)} \in \mathbb{R}^{S \times d}$  (rank  $R = S$ ), storing  $2dS$  floats per segment instead of  $d^2$  and computing the readout  $M^{(i)} q = U^{(i)\top} (W^{(i)} q)$  at  $2dS$  cost instead of  $d^2$ . The factorization is exact for a rank- $\leq S$  matrix, so the model output is unchanged up to floating-point error—`test_factored_equals_dense` verifies the identity (threshold  $10^{-4}$ ; measured gap  $< 5 \times 10^{-7}$ )—and at the official Zoology task (512, 64) the factored model matches the dense one within seed noise (dense  $99.02 \pm 0.17$ , factored  $99.18 \pm 0.08$ , 3 seeds). Factoring removes the quadratic-in-width term ( $d^2 \rightarrow 2dS$ : 4096  $\rightarrow$  2048 elements at  $d = 64$ ), but the state still grows with the number of segments,  $O(T/S)$ .

**Capped memory.** We bound the number of *live* cached segments at  $K_{\max}$  and compress the overflow into a single low-rank background of rank  $r_{\text{bg}}$ . The recurrent state becomes

$$\text{state} = K_{\max} \cdot \text{per\_seg} + (2d r_{\text{bg}} + d + m) + K_{\max},$$

independent of  $T$ , where  $\text{per\_seg} = 2dS + d + m$ . Compression is a deterministic FIFO eviction with normalization of the compressed readout; live segments and the compressed background are mixed inside IGA’s single global ratio, so the interference-gated aggregation of §3.1 is preserved. At the official task (512, 64) the context is exactly  $K_{\max} = 32$  segments, so no eviction occurs and the capped model is a no-op: recall is identical to the uncapped factored model—99.23/99.30/99.02 (mean 99.18) across three seeds. The honest price of constant memory is that segments older than  $K_{\max}$  are compressed (lossy): recall of facts within the last  $K_{\max}$  segments tracks the unlimited baseline even with hundreds of evictions, while older facts are not individually recoverable—the explicit recall/memory tradeoff.

## 4 Experiments—associative recall

All numbers are mean  $\pm$  std over 3 seeds unless noted. Baselines: exact attention (2-layer MHA with learned positions), Based (official hybrid: BaseConv + Taylor-exp linear attention), Mamba (S6; pure-PyTorch reimplementation with exact parallel scan, unit-tested against the sequential reference). Backbone, initialization (GPT-2 scheme) and hyperparameters follow the Zoology Figure-2 configuration. A negative finding, verified against the *official* implementation: the 2-layer attention-only Figure-2 configuration does not form the induction circuit under our infinite-stream protocol (fresh samples every step remove the memorization bootstrap); it plateaus at  $\sim 13\%$  across learning rates, and the official `zoology.model.LanguageModel` trained under the same loop gives 12.3%—identical to our replica—while under the official fixed-data regime our replica *does* reproduce the published induction-head transition. We therefore report **attention-conv**—a single-layer exact-attention model with the same short-convolution stem every other model has—as the working upper anchor.

### 4.1 Verbatim MQAR (our protocol)

Train at  $n = 48$  ( $n_k = 192$ ), evaluate without retraining; chance 6.25%.

| model                    | $n=48$         | $n=64$         | $n=96$         | $n=128$                          |
|--------------------------|----------------|----------------|----------------|----------------------------------|
| attention (Zoology fig2) | $12.9 \pm 0.0$ | $11.1 \pm 0.5$ | $9.7 \pm 0.5$  | $9.0 \pm 0.4$                    |
| based                    | $8.3 \pm 0.1$  | $7.9 \pm 0.2$  | $7.3 \pm 0.2$  | $7.3 \pm 0.1$                    |
| mamba                    | $13.3 \pm 0.0$ | $12.3 \pm 0.1$ | $10.6 \pm 0.2$ | $9.6 \pm 0.4$                    |
| FIXA(64)                 | $77.9 \pm 1.0$ | $71.6 \pm 0.8$ | $59.9 \pm 0.4$ | $50.9 \pm 0.4$                   |
| FIXA(128)                | $92.1 \pm 0.5$ | $88.8 \pm 0.6$ | $81.8 \pm 1.0$ | $74.5 \pm 1.1$                   |
| IGA-hist (IDs)           | $98.3 \pm 0.1$ | $98.3 \pm 0.1$ | $98.4 \pm 0.2$ | <b><math>98.5 \pm 0.1</math></b> |
| IGA-sketch (IDs)         | $98.1 \pm 0.1$ | $98.1 \pm 0.1$ | $98.1 \pm 0.2$ | $98.0 \pm 0.2$                   |
| <b>IGA-poly (no IDs)</b> | $97.7 \pm 0.1$ | $97.6 \pm 0.2$ | $97.5 \pm 0.2$ | <b><math>97.1 \pm 0.2</math></b> |

All IGA variants are flat in evaluation length; the ID-free polynomial router pays 0.8–1.4 points of sketch noise relative to the exact histogram. The 2-layer literature baselines fail in this regime (48 pairs at  $d = 64$ , far beyond their capacity-favorable setting).

## 4.2 MQAR-indirect

Same protocol; the query never repeats the key.

| model                                  | $n=48$          | $n=64$          | $n=96$          | $n=128$                          |
|----------------------------------------|-----------------|-----------------|-----------------|----------------------------------|
| attention (Zoology fig2)               | $13.1 \pm 0.1$  | $11.5 \pm 0.4$  | $9.9 \pm 0.2$   | $9.2 \pm 0.2$                    |
| based                                  | $7.4 \pm 0.3$   | $7.3 \pm 0.1$   | $6.8 \pm 0.2$   | $6.8 \pm 0.1$                    |
| mamba                                  | $13.3 \pm 0.0$  | $12.2 \pm 0.1$  | $10.6 \pm 0.3$  | $9.6 \pm 0.3$                    |
| FIXA(64)                               | $57.6 \pm 20.8$ | $51.4 \pm 19.4$ | $42.5 \pm 16.2$ | $36.8 \pm 13.9$                  |
| IGA-sketch (control)                   | $57.2 \pm 20.6$ | $50.9 \pm 19.1$ | $42.1 \pm 15.9$ | $36.4 \pm 13.6$                  |
| IGA-emb+aux (exp family)               | $68.6 \pm 17.8$ | $62.2 \pm 17.9$ | $52.1 \pm 16.0$ | $45.1 \pm 13.8$                  |
| <b>IGA-poly+aux (privileged)</b>       | $96.1 \pm 0.7$  | $95.8 \pm 0.7$  | $95.4 \pm 0.7$  | <b><math>94.8 \pm 0.9</math></b> |
| <b>IGA-poly distilled (label-free)</b> | $93.5 \pm 1.0$  | $92.6 \pm 1.4$  | $90.5 \pm 1.8$  | <b><math>88.2 \pm 2.2</math></b> |

The ID-counting control tracks FIXA seed-by-seed (routing 6.0% = chance), confirming the benchmark removes the verbatim shortcut by construction. The polynomial-kernel router preserves the flat-recall property with *fully learned* identity: +58 points over the fixed memory at  $n = 128$ , and the label-free distilled variant retains 93–96% of the privileged upper bound.

## 4.3 Official Zoology protocol (vocabulary 8192)

Official generator and training loop; 3 seeds, mean  $\pm$  std.

| model                                    | (128, 8) easy  | (512, 64) hard                   |
|------------------------------------------|----------------|----------------------------------|
| attention (full softmax)                 | $99.4 \pm 0.3$ | $100.0 \pm 0.0$                  |
| attention-conv (1-layer anchor)          | $99.3 \pm 0.3$ | $99.4 \pm 0.0$                   |
| based                                    | $99.2 \pm 0.2$ | $79.5 \pm 1.4$                   |
| mamba                                    | $99.1 \pm 0.0$ | $23.2 \pm 14.0$                  |
| <b>IGA-sketch (ours)</b>                 | $99.1 \pm 0.0$ | <b><math>99.1 \pm 0.0</math></b> |
| <b>IGA-poly (ours, learned identity)</b> | $99.1 \pm 0.0$ | <b><math>99.2 \pm 0.1</math></b> |

The discriminating regime is (512, 64): 64 key-value pairs over a 512-token context (32 segments of  $S = 16$ — $2\times$  more segments than any setting above). All six models solve the easy task. On the hard task the efficient baselines separate exactly as the Based paper reports: **Mamba collapses** ( $23.2 \pm 14.0$ , **one seed at 7%—the fixed-state saturation**) and **Based degrades** ( $79.5 \pm 1.4$ ). **Both IGA models hold at  $\sim 99\%$** , matching full softmax attention (100%) and the convolutional attention anchor (99.4), at sub-quadratic per-token cost. This is the citable result that closes the comparison: on the literature’s own benchmark, in its hardest standard setting, the growing memory keeps near-perfect recall where the fixed-state models fail.

#### 4.4 Inference cost

Batch 1,  $d = 64$ , fp32, reference Python loops on both sides (scaling comparison, not engineering constants).

| model                          | $T=4096$<br>tok/s | $T=16384$<br>tok/s | $T=65536$<br>tok/s | state @ $T=65536$       |
|--------------------------------|-------------------|--------------------|--------------------|-------------------------|
| attention (SDPA, KV cache)     | 3577              | 997                | 253                | 32.0 MB ( $\propto T$ ) |
| IGA-poly recurrent, soft gate  | 3337              | 3721               | 3167               | <b>0.39 MB</b> (const)  |
| IGA-poly recurrent, hard top-1 | 4885              | 4725               | <b>4455</b>        | <b>0.39 MB</b> (const)  |

IGA throughput is  $\sim$ flat in context length; incremental exact attention degrades linearly,  $17.6\times$  slower at 64k. The state column reports the constant-memory state of  $\$3.3$ —**0.39 MB** constant—against a KV cache that grows  $\propto T$  (32 MB at 64k). The throughput crossover and the constant sub-KV state are the claims.

#### 4.5 Constant-memory scaling and the long-context regime

With the factored + capped state, the recurrent state is constant in  $T$ . Measured with `state_bytes` ( $d = 64$ ,  $K_{\max} = 32$ ,  $r_{\text{bg}} = 16$ , fp32) at 0.39 MB against the KV cache  $2Td$ :

| $T$   | KV cache | IGA (constant) | ratio         |
|-------|----------|----------------|---------------|
| 512   | 0.25 MB  | 0.39 MB        | $1.53\times$  |
| 1024  | 0.50 MB  | 0.39 MB        | $0.76\times$  |
| 2048  | 1.00 MB  | 0.39 MB        | $0.38\times$  |
| 4096  | 2.00 MB  | 0.39 MB        | $0.19\times$  |
| 65536 | 32.0 MB  | 0.39 MB        | $0.012\times$ |

The state equals the cache at the crossover  $T \approx 2K_{\max}S$  (where the cache fills exactly  $K_{\max}$  segments); below it the constant state is larger—it is sized for capacity, not the current context—and above it the advantage grows without bound. Projected to a 1T-parameter regime ( $d = 8192$ ,  $K_{\max} = 4096$ ,  $r_{\text{bg}} = 4$ ), the constant state is 209.5 GB =  $0.065\times$  the 3.1 TB KV cache—**15 $\times$  smaller, and constant**—where the dense IGA state would be 800 TB ( $256\times$  larger) and even the factored-but-uncapped state is 3.2 TB (parity, still growing). The constant sub-KV state is the memory result.

## 5 Real-text language modeling: constant memory at no perplexity cost

All §4 results use synthetic associative recall. We close the remaining question—does the constant-memory IGA carry a perplexity penalty on *real* text?—by training a 2-layer,  $d = 128$  IGA language model (segment size 64; tensor-sketch polynomial router reading raw tokens; factored state of §3.3) against the *identical* architecture with the gate disabled (`gated=False`, the FIXA fixed-memory baseline of the same width and parameter count). Data is character-level TinyStories, 88 MB, downloaded—no synthetic data is generated for any language-model experiment. Optimizer AdamW  $3 \times 10^{-4}$ , batch 16, 4000 steps, 3 seeds, contexts  $\{1024, 2048\}$ . We report bits-per-character (bpc) on a 5% held out split and a planted-fact recall probe—the MQAR analogue on natural text—at three distances (ctx/4, ctx/2, ctx−64).

| ctx  | IGA bpc             | FIXA bpc            | $\Delta$ bpc (IGA−FIXA) | recall@ctx−64 IGA | recall@ctx−64 FIXA |
|------|---------------------|---------------------|-------------------------|-------------------|--------------------|
| 1024 | $2.1607 \pm 0.0060$ | $2.1621 \pm 0.0084$ | −0.0015                 | 1.67              | 1.67               |
| 2048 | $2.1138 \pm 0.0114$ | $2.1143 \pm 0.0127$ | −0.0005                 | 2.29              | 2.08               |

Two findings. (i) **No perplexity cost.** The bpc difference is −0.0015 (ctx 1024) and −0.0005 (ctx 2048), both an order of magnitude below the per-seed std (0.006–0.013); IGA is never consistently worse across the six (model, seed) runs. The constant-memory state therefore does not degrade language modeling relative to a fixed-memory model of the same width and parameter count. (ii) **The recall advantage is task-specific.** The planted-fact probe is near chance for both models at both contexts (1.7–2.3%, per-seed spread 0–4.4%—pure noise, no distance dependence); the flat-recall advantage of §4 does not transfer to retrieval of an arbitrary planted fact in natural text at  $d = 128$ . The structured MQAR benchmark remains the right instrument for the associative-recall claim; natural-text retrieval at this scale does not separate the architectures, and scaling  $d/\text{ctx}$  to surface a separation is left to future work (GPU-budget-bound).

**Compute.** IGA costs  $5.2 \times$  (ctx 1024) and  $4.9 \times$  (ctx 2048) more wall-clock per step than FIXA in our reference Python-loop implementation—the gate’s per-segment normalization, dominated by loop overhead rather than asymptotic FLOPs. The tradeoff is memory-vs-compute (constant  $O(1)$  state against higher per-step compute), not memory-vs-quality: quality is matched within noise.

We note honestly that a preliminary 300-step run in the parent project reported a  $\sim 0.016$  bpc gain at  $\text{ctx} \geq 2048$ ; it does **not** reproduce under the 3-seed, 4000-step protocol here ( $\Delta = -0.0005$ , within noise). The language-model result is a no-regression result, not a quality gain.

## 6 Deployment economics: where constant memory pays

This section synthesizes §4.4 (throughput), §4.5 (memory) and §5 (LM quality) into the deployment-cost picture—the regime in which a growing-memory model is actually meant to replace a KV-cache transformer.

**Training (one-time).** Per step, softmax attention is  $O(T^2d)$ —the  $QK^\top$  and AV products over the full  $T \times T$  matrix—while IGA’s linear-attention recurrence is  $O(Td^2)$  (the factored state update is  $O(TdS)$ , smaller). The asymptotic FLOP scaling therefore favors IGA as context grows, with ratio  $T/d$ :  $\sim 8 \times$  fewer attention-FLOPs at ctx 1024 ( $d = 128$ ),  $\sim 512 \times$  at  $T = 64k$  ( $d = 64$ ). We did *not*



train a transformer language-model baseline, so we state the training-cost comparison as complexity analysis, not as a measured number; the one comparison we *do* measure end-to-end is inference throughput against exact attention (§4.4: 14–17.6× at 64k). The 5× wall-clock premium of §5, in contrast, *is* measured—against the *fixed-memory linear baseline* (FIXA), the relevant same-family control for the gate’s overhead.

**Inference (recurring, per generated token).** A KV-cache transformer reads the full cache per token— $O(TdL)$  memory traffic—and holds a cache of  $2TdL$  that grows  $\propto T$ . IGA reads a constant  $O(K_{\max}dS)$  per token and holds a constant state. Measured at batch 1,  $d = 64$  (§4.4): IGA holds  $\sim 4.5\text{k tok/s}$  flat from  $T = 4\text{k}$  to 64k, while incremental exact attention degrades  $\sim 14\times$  to 253 tok/s at 64k. Per-token inference cost of IGA is constant in  $T$ ; the transformer’s grows with  $T$ .

**Amortization.** Training is paid once; inference is paid per query across the model’s serving lifetime. The one-time 5× gate premium (against the linear baseline) is amortized by the recurring inference savings once the served context is long enough that the constant state undercuts the cache—the crossover  $T \approx 2K_{\max}S$  of §4.5, 15× smaller projected to 1T. Below the crossover, where the cache is itself small and attention is fast, the transformer is the cheaper option and IGA’s advantage is only the recall capacity of §4; above it, IGA’s per-token cost stops growing while the transformer’s keeps growing, so the total cost of serving long contexts eventually favors IGA by an unbounded margin. Past the point where the KV cache no longer fits on device, IGA is not the cheaper option but the only feasible one.

This is a cost-structure argument, not a measured deployment dollar figure: absolute cost depends on hardware, batching and kernel fusion not modeled here (reference Python loops). The structural claim— $O(1)$  memory and  $O(1)$ -in- $T$  per-token compute against  $O(T)$  and  $O(T)$  for the cache, with matched LM quality (§5)—is what the constant-memory result buys. That is the “no pain”: the growing-memory cost is removed. The “same gain” is the matched recall (§4.3) and matched perplexity (§5).

## 7 Related work

**Efficient-attention recall benchmarks.** Zoology [Arora et al., 2023] established MQAR as the discriminating task between attention and efficient mixers; Based [Arora et al., 2024] frames the recall-throughput tradeoff. We adopt their harness and add MQAR-indirect, which isolates *identity learning* from *identity exploitation*—to our knowledge the first recall benchmark where queries never repeat keys by construction.

**Growing memory.** Memory Caching [Behrouz et al., 2026] proposes segment-cached memories with gated aggregation (GRM among its variants). We keep its state layout ( $M^{(i)}, z^{(i)}$ ) and cost, and replace its aggregation (the collapse cause) and its routing (the saturation cause). Its state grows with the number of cached segments ( $O(T/S)$ , quadratic in width via  $d^2$ ); our factored + capped state (§3.3) is constant in  $T$  and linear in width, removing the growth that would otherwise exceed the KV cache it is meant to replace.

**Polynomial sketches in attention.** PolySketchFormer [Kacham et al., 2023] uses polynomial sketches to *replace the attention kernel itself*. Our use is different: the sketch summarizes *segment identity* for routing in a growing-memory RNN; attention/readout keeps its own feature map. Count sketch [Charikar et al., 2002], count-min [Cormode and Muthukrishnan, 2005] and tensor sketch

[Pham and Pagh, 2013] supply the estimators; the median-of- $J$  variant mirrors count-min’s min under signed estimates.

**Linear attention and SSMs.** The fixed memory is Katharopoulos et al. [2020]; Based and Mamba are the strongest small-scale recall baselines in this family. Performer [Choromanski et al., 2021] approximates  $\exp(q \cdot k)$  with random features; our variance argument (§3.2) explains why that family cannot provide  $\delta$ -sharp identity routing at fixed feature budget.

**Distillation across compute classes.** SMT [Kumar and Isola, 2026] trains an RNN with a parallel teacher; our teacher-to-router distillation transfers *routing* (not state updates) and is, to our knowledge, the first label-free recovery of identity routing in a growing-memory model.

## 8 Limitations and honest notes

- Single-layer  $d = 64$  synthetic recall is the headline setting; the language model of §5 uses 2 layers,  $d = 128$ , char-level TinyStories, 3 seeds, two contexts (downloaded corpus, no synthetic data). IGA matches the fixed-memory baseline in bpc within seed noise (no perplexity cost), but the planted-fact recall probe is near chance for both at this scale, and the  $\sim 0.016$  bpc gain reported by a preliminary 300-step run does *not* reproduce under the 3-seed 4000-step protocol; the associative-recall advantage is demonstrated on structured MQAR (§4), not on natural-text retrieval at  $d = 128$ . Scaling  $d$  and context to surface a recall separation on real text is left to future work (GPU-budget-bound).
- The indirect bijection is 1-to-1 and deterministic; real paraphrase is many-to-many and noisy.
- The privileged-aux arm uses bijection labels; the distillation arm removes them but adds a quadratic teacher *during training only*. No arm learns routing from the recall gradient alone—the routing-family analysis of §3.2 suggests this is impossible in this regime (perverse equilibrium); we state it as a conjecture, not a theorem.
- Mamba is a pure-PyTorch reimplementation (exact scan, unit-tested); absolute wall-clock comparisons to fused-kernel implementations would be unfair and are not made—only task accuracy is compared.
- The cost benchmark uses reference Python loops on both sides (no fused kernels); it measures *scaling*, not engineering constants. The recurrent state is constant in  $T$  (§3.3) and below the KV cache from  $T \approx 1\text{k}$  onward; at the smallest contexts ( $T \lesssim 2K_{\max}S$ ) the constant state is sized for capacity and exceeds the cache. Segments older than  $K_{\max}$  are compressed (lossy), the explicit recall/memory tradeoff.
- Seeds: 3 per cell. Std is sample std (Bessel). No hyperparameter search was performed for our models beyond the E0 microbenchmark ( $p, m, J, r$ ); baselines received the official learning-rate sweep.

## 9 Reproduction

```

pip install -r requirements.txt
python3 tests/test_sanity.py          # 14 equivalence/property tests
                                     # (incl. factored==dense <5e-7, IGA==fixed at init)

```

```

python3 scripts/run_cost.py                # inference cost (~5 min)
python3 scripts/run_multiseed.py all        # our models, 3 seeds
python3 scripts/run_baselines_ours.py all   # literature baselines, our protocol
python3 scripts/run_zoology.py all         # official protocol (incl. (512,64))
python3 scripts/aggregate.py               # tables (mean +/- std)
python3 scripts/run_factored_check.py      # factored vs dense recall at (512,64), 3 seeds
python3 scripts/run_fixb_check.py          # capped-memory recall == factored + constant-state table
python3 scripts/project_memory.py          # memory projection (small/medium/1T)
# Real-text language modeling (downloaded TinyStories; isolated folder):
python3 multicamada/run_lm_ml.py --steps 4000 --ctx 1024 --seeds 0,1,2 --tag lm_full
python3 multicamada/run_lm_ml.py --steps 4000 --ctx 2048 --seeds 0,1,2 --tag lm_full_ctx2048

```

The language-model experiment uses the repository’s existing multi-layer model IGALM (`src/models/iga_layer.py`,  $n_{\text{layers}}=2$ ,  $d=128$ , factored-capable) compared against itself with `gated=False` (the FIXA baseline of matched width/parameters), isolated in the `multicamada/` folder. The corpus is the downloaded TinyStories sample (88 MB); no synthetic LM data is generated. All runs are resumable (skip-if-exists per JSON in `results/raw/`).

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